CERTAIN-COAST - concept note

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast

NATO Science for Peace and Security (SPS) Programme - Multi-Year Project, call 2027-1 - 36 months - about EUR 382,500 requested from NATO - coordinator: Faculty of Geology and Mining, Polytechnic University of Tirana, Albania

1. In short

On Albania's Adriatic coast, the drinking-water wells and port of Durrës and the landfall of a gas pipeline to Italy near Fier stand on low ground, which at Durrës is sinking, close to active faults. Three dangers threaten them: seawater entering the wells, ground that turns almost liquid in an earthquake, and the sea flooding the land in storms. The dangers are linked, because sinking land and rising seas worsen all three and pumping that protects against one can increase another, but today they are watched separately and no warning says how often it can be wrong.

CERTAIN-COAST will watch them together. Satellites, GNSS and new ground sensors will feed one computer model of the coast, copied by a fast artificial-intelligence model whose error is stated for every forecast. Each warning will state in advance how often it may miss a dangerous event, and this will be checked in twelve months of live operation with the Ministry of Infrastructure and Energy, which will use the warnings to protect critical infrastructure. The project will show how many years a safe way of operating the three assets remains, and plan safe pumping tests that would sharpen that estimate.

2. The security challenge

Three kinds of critical infrastructure of a NATO Ally stand on the same low Adriatic coastal plain of Albania, at two sites. At Durrës lie the wellfields that supply drinking water and the Port of Durrës, from which 93.6% of Albania's seaborne freight left in the second quarter of 2025 (INSTAT). Further south, near Fier, the Trans Adriatic Pipeline (TAP) leaves the shore for its 105 km crossing to Italy, carrying gas from Azerbaijan to Europe. The plain is built of soft sediments that settle slowly (Métois et al., 2020); at Durrës the ground is sinking close to active thrust faults, and the Mw 6.4 earthquake of 26 November 2019 caused liquefaction along the coast.

The assets face three linked hazards. Seawater intrudes into the aquifer and can make wells unusable for years. In an earthquake, saturated sand can liquefy under quays, foundations and the landfall. Storm surges and extreme sea levels flood low ground. Land subsidence and sea-level rise worsen all three at once, because they raise the sea and the water table together. Pumping moves the hazards in opposite directions: it draws seawater towards the wells but lowers the water table and so strengthens the ground against liquefaction. Managing one hazard alone can increase another.

Today these hazards are monitored by different institutions with different models, and no warning states how often it misses. For security decisions this is the critical gap: whoever must cut pumping, restrict a quay or inspect the landfall needs to know how far a warning can be trusted, and the national authorities need one consistent picture of water, port and energy after an earthquake or a storm.

The security challenge is therefore to keep three essential services of an Ally working, and to know in advance when they will stop being safe. The project serves three of NATO's seven baseline requirements for national resilience (resilient food and water resources, resilient civil transportation systems and resilient energy supplies) and the topic of particular interest of the 2027-1 call, "protection of critical infrastructure, supplies and personnel; maritime and undersea infrastructure resilience". NATO's 2022 Strategic Concept calls climate change "a crisis and threat multiplier"; on this coast the multiplication is physical and measurable.

A second challenge is trust in machine-learned tools used in security decisions. CERTAIN-COAST uses artificial intelligence only with a certified error bound and a miss rate that is checked in operation, in line with NATO's Principles of Responsible Use of AI.

3. Who uses it, and how

Link to security. The project protects three essential services of a NATO Ally (drinking water, a port that handles most of the country's seaborne freight, and the landfall of the Trans Adriatic Pipeline near Fier) and serves three of NATO's seven baseline requirements for national resilience: resilient food and water resources, resilient civil transportation systems and resilient energy supplies, which also

underpin civil support to the military. It falls under the 2027-1 topic "protection of critical infrastructure, supplies and personnel; maritime and undersea infrastructure resilience": the TAP landfall is where an undersea gas pipeline between two Allies, Albania and Italy, leaves the shore.

Use cases:

- Ministry of Infrastructure and Energy of the Republic of Albania, the end-user: receives the bulletins for the wellfields, the port and the landfall during the live operation, agrees their warning thresholds, and uses them in its planning for the water supply, the energy and transport infrastructure and the geological hazards under its responsibility, for example to set priorities after an earthquake or a storm surge (hours to days ahead) and to plan for a loss of drinking-water wells (weeks to months ahead); staff time in kind.
- The authorities that issue water-use permits and the water supply: a chloride early warning and a pumping and drought rule, offered as a policy brief, so that pumping at the wellfields can be cut or shifted before chloride reaches drinking-water wells, safely even while the source of the chloride is unknown.
- The Port of Durrës and the TAP landfall near Fier: warnings of ground motion, liquefaction and storm surges, computed from satellite radar, GNSS and public data, to set inspection priorities after an earthquake and prepare for surges (hours to days ahead); no data or access from the asset owners is needed.
- National authorities and resilience planners: the certified year from which no operating plan keeps the three assets safe together, and the sensors and measures that delay it most.

Each bulletin states its certified miss rate, so decision-makers know how far it can be trusted. The method uses free satellite data and transfers to other subsiding coasts of Allies and partners with wellfields, ports and energy landfalls.

4. What the project will deliver, and what is new

By Month 36 the project delivers:

1. A coastal motion atlas from Durrës to Fier and relative sea level at each asset, from Sentinel-1 InSAR, reflectors and GNSS (D1.1).
2. A monitoring network chosen for the warnings, in the project's own boreholes on public land (D4.2): piezometers with conductivity loggers, a barometer and sealed hub channels that correct them, a rain gauge, GNSS, seismic nodes and, where access allows, fibre-optic sensing.
3. One calibrated model that couples groundwater flow and salt, aquifer compaction, liquefaction under a moving water table and coastal water levels, its inland boundary learned from the heads (D2.1), and a fast neural-operator copy with a certified error bound (D2.2).
4. Certified alarms for seawater intrusion, liquefaction and extreme water levels whose miss rate is bounded in advance (indicatively 5–10%), with a joint guarantee for the three assets, back-tested since 2014 under a pre-registered protocol (D3.1, D3.2).
5. Decisions for the wellfields: the effect of each wellfield's pumping on salinity with the hidden drought demand removed; the salinity under drier recharge; the certified safe operating window and the year it closes; safe pumping tests planned to sharpen it; a pumping rule; and pumping safe while the source of the chloride is unknown.
6. Twelve months of live bulletins with the end-user, every outcome logged and the miss rates verified (D5.1); operating procedures and a policy brief (D6.2); the model and the alarm algorithm, built by the NPD, released as open-source MATLAB code (https://github.com/dulianzeqiraj/certain-coast) with open data and a DOI (D6.1).

How this advances the state of the art. The alarms apply conformal and anytime-valid methods of 2023–2026 to coupled coastal hazards. Nine new contributions were tested before the project. (i) Guarantees for quantities nobody measures: inputs and boundaries known only through rain, with bands valid when the simulator or rain product is wrong. (ii) The value of a sensor as the guaranteed band it removes and the years of safe operation it adds. (iii) Pumping effects free of hidden confounding, read from the heads. (iv) The certified safe operating window and safe pumping tests chosen for the years they add. (v) A network that corrects its own sensors for air pressure and logger

drift. (vi) Warnings between samples: a certified band for the chance that chloride crosses its limit at any time in the season (34 times the exceedance on sampling dates in tests). (vii) Decisions before the source of the salt is known: pumping safe for every source the tracers cannot exclude (in tests at most 6.9% risk for a 10% budget; assuming seawater intrusion: 98%). (viii) A boundary that learns the aquifer beyond the model: exact for any linear aquifer, stable by construction, learned from two piezometers (6 mm against 372 mm on an unseen drought in tests). (ix) Which layer feels an earthquake, and its step as a free pumping test (in tests the usual estimate overstated the soft-sand step 35 times). The integrated system reaches TRL 6 on the project's network at Durrës.

5. How the project works

Seven work packages run over 36 months in three phases, separated by evidence gates with thresholds fixed in advance.

Observe and measure (Months 1–12). WP4 designs the network before drilling or buying it: each candidate sensor (conductivity loggers, a rain gauge, GNSS, seismic nodes, a barometer and hub reference channels) is ranked by the guaranteed warning band it removes and the years of safe operation it adds per euro; the network is chosen against the worst plausible sensor error, with a certified optimality gap (D4.1, Month 6). WP1 obtains vertical land motion and relative sea level at each asset from Sentinel-1 InSAR since 2014 and GNSS, and the head limit for pumping from aquifer compaction.

Model and certify (Months 4–30). WP2 builds one coupled model in stages (flow and salt, then compaction and liquefaction). The inputs and boundaries nobody measures are reconstructed from rain and heads by a generative prior with an ensemble update. Their bands, calibrated on simulated aquifers and measured anchor periods, stay valid if the simulator is wrong or the climate drifts. The inland boundary is learned from two piezometers as passive storage compartments. Gate MS3 (Month 18) requires held-out heads, chloride, InSAR and seismic velocity inside the 90% intervals. A neural-operator copy with a certified error bound then serves the warnings. WP3 agrees thresholds and tolerable miss rates with the end-user, builds the alarms (crossings between samples included), back-tests them since 2014 under a protocol registered before the results; gates MS4 (Month 22) and MS5 (Month 27) decide live operation and confirm the miss rates.

Operate, verify and hand over (Months 22–36). WP5 runs the system live for twelve months, logs every bulletin with its outcome, verifies the miss rates with anytime-valid tests and runs a post-event protocol. It estimates what each wellfield's pumping does to the salinity with the hidden drought demand removed, reading the recharge from the heads under a calibrated check. While the source of the chloride is uncertain, tracers chosen for the decisions they change keep only the sources the data cannot exclude, updated monthly, and the pumping stays safe for all of them. It computes the salinity under drier recharge, the certified safe operating window, its closing year and safe pumping tests that would sharpen it, and prepares a pumping rule for the water-use permits. WP6 trains early-career researchers and the end-user's staff, releases the open-source code and transfers the method to a partner coast. WP7 manages data access, security classification, risks and reporting.

Collaboration. The NPD builds the coupled model and the alarm algorithm and releases them as open-source MATLAB/Octave code; the Albanian team leads WP1–WP5 and WP7; Qatar University leads WP6, a transfer study on the east coast of Qatar and a user-side review of the bulletins. Early-career researchers of both countries work across work packages, with two-month exchanges. The end-user agrees the thresholds and reviews the bulletins; a steering committee with it meets every six months, and each gate has a named fallback.

6. The end-user: what it gives and what it gains

End-user	Contribution (in kind; no SPS funds)	What it gains
Ministry of Infrastructure and Energy of the Republic of Albania (Ministria e Infrastrukturës dhe Energjisë): the end-user	A contact person and a small team on the steering committee; agrees the warning thresholds of the bulletins it uses; reviews the bulletins during the live operation and takes part in the final workshop.	Warnings with a stated miss rate for the wellfields, the port and the landfall, for its planning for the water supply, the energy and transport infrastructure and the geological hazards under its responsibility; post-event reports; trained staff, the software and the operating procedures at the hand-over; no procurement, ownership or maintenance obligations.

The Port of Durrës and the TAP landfall near Fier are assessed only from satellite radar, GNSS and public data; no data or access is needed from their owners. The monitoring network is drilled by the project on public land and stays in Albania with the Polytechnic University of Tirana. The bulletins are advisory: operational decisions stay with the operators, and any warning to the public is issued only by the competent authorities. Data shared with the project are restricted, used under data agreements, and every publication is reviewed for sensitive content before release; software and non-restricted data products are open, with a DOI.

7. Work plan and milestones

Work package	Title	Duration
WP1	Coastal motion and relative sea level	Months 1-36
WP2	One coupled model of the coast and its certified fast copy	Months 4-24
WP3	Certified multi-hazard alarm	Months 13-34
WP4	Monitoring network designed for the warnings	Months 1-36
WP5	Live demonstration at the wellfields, port and landfall	Months 22-36
WP6	Training, transfer and dissemination	Months 1-36
WP7	Management, data and risk	Months 1-36

Milestone	Month	What is achieved
MS1	6	Network design agreed
MS2	12	Network streaming; coastal motion known
MS3	18	Coupled model calibrated
MS4	22	Go/no-go for live operation
MS5	27	Miss-rate targets confirmed in back-testing
MS6	34	Twelve months of live operation completed
MS7	36	Hand-over to the end-user

8. Budget (EUR)

Category	EUR
Equipment	182,800
Training	41,000
Communication and Publications	8,000
Travel	48,000
Consumables	22,700
Other	32,000
Stipends	48,000
Total requested from NATO SPS	382,500

Salaries, overheads and the in-kind contributions of the institutions and the end-user are not charged to NATO. The monitoring network bought by the project stays in Albania and keeps the bulletins running after the project.

9. Evidence from the preparatory work (2026)

- Boundaries known only through rain: on 500 simulated aquifers with four loggers, the raw ensemble missed 17–20% of the weeks; the calibrated bands met the 10% target (0.095–0.100).
- Pumping and hidden drought demand: in a test bed with eight wells, the summed effect of pumping on salinity (true value 0.94) was overstated by 0.59 without adjustment, by 0.56 with a substitute built from the pumping history (as in the Time Series Deconfounder) and by 0.00 with the recharge read from the heads; the physics check had size 6% and power 99–100%.

- Alarms and operating window: an alarm with an extreme-aware margin kept the share of seasons above the chloride limit at 4–9% across storylines, where a static alarm allowed 13–21%; the certified safe operating window closed after 38 years, while plans that watch one hazard believed they had 43–97 years; safe pumping tests chosen for the years they add raised the certified closing year from 10.8 to 31.9 years (true value 45) with chloride below 123 mg/L in 95% of runs; the anytime-valid test wrongly flagged a correctly calibrated alarm in 2.2% of runs, against 68% for re-checking a 95% bound after every event.
- Sensors that interfere: a 12-sensor network with a barometer and hub reference channels kept 52% more worst-case information than the classical design, certified within 18% of the best; ignoring the interference let 90% intervals cover only 49–62% of cases, the correction 88–93%.
- Crossings between samples: chloride with an 8-day time scale crossed 250 mg/L within a season with probability 0.78, 34 times the exceedance on a sampling date; quarterly samples left this anywhere in 0.18–1, a daily logger in 0.76–0.80; a conformal margin restored 90.3% coverage for skewed chloride (74% without).

These are tests on synthetic coasts and on the existing chloride model with illustrative values, not results at Durrës; the project calibrates and verifies them on site.

10. Contact

Assoc. Prof. Dr. Dulian Zeqiraj, NATO-country Project Director - Department of Energy Resources, Faculty of Geology and Mining, Polytechnic University of Tirana, Albania - dulian.zeqiraj@fgjm.edu.al - open-source MATLAB code: https://github.com/dulianzeqiraj/certain-coast